Introduction to Robotics: Introduction to leJOS

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Lecture Plan

- Java Compatibility
- Control Center
- leJOS APIs
- leJOS Examples

Java Compatibility 1

- leJOS
  - lejos is similar to legos, except there is a “J” for Java
  - lejos means “far” in Spanish
  - lego Java Operating System
    - A customized minimal Java Virtual Machine (JVM)
- Imagine the electronic devices about 15 years ago, when leJOS was created
  - 12K memory
  - 2G network
  - 0.3 megapixel camera
  - Bluetooth 1.2
  - Price: $400 for Black Berry Pearl
    $600 for 1st gen iPhone

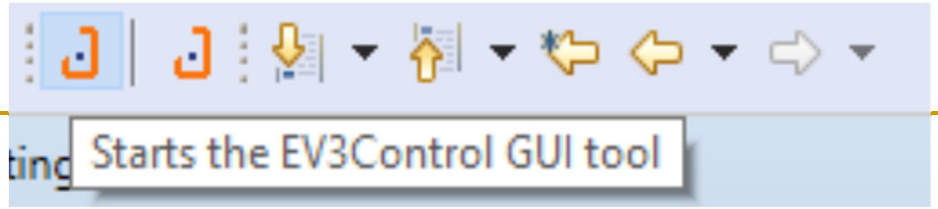
Java Compatibility 2

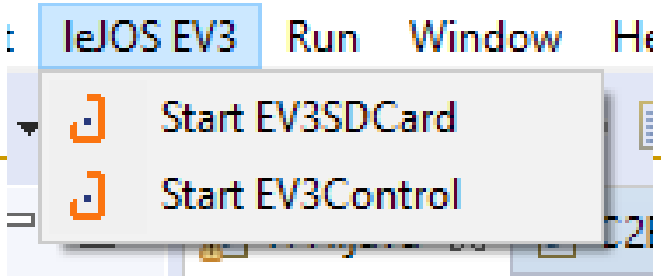
- leJOS for EV3
  - Now it runs on Linux
  - With larger memory: e.g. with a Micro-SD card
    - If you want to create your own SD card, please make sure the partition is not more than 8GB: to work with the FAT 32 file system
  - We can use a fully featured JVM
  - Homemade JVM can be faster on EV3:
    - e.g. 20 us for a sensor reading v.s. 8 ms on a Segway robot
  - But the EV3 Linux is slow
    - You need to wait more than 20s to boot leJOS on EV3

A Segway Scooter

Control Center 1

- The Control Center is a comprehensive utility console, to help controlling and configuring of EV3, from the PC
- To run the control center:
  - In Eclipse:
    - Click the “Starts the EV3Control GUI tool” icon in the quick access toolbar
    - Select “Start EV3Control” option in the “leJOS EV3 menu”
  - In your file system: run “ev3control.bat” from the leJOS bin directory.





Control Center 2

- Once the Control Center is opened
  - Search and connect EV3 bricks around
  - Stop running program on Ev3
  - Shutdown EV3
  - Control motors, read sensor
  - Change EV3 settings
  - …

Control Center 3

- Programs: EV3 File Browser

EV3 Control Center

Name	Protocol	Address	Status
EV3-21		192.168.0.21	CONNECTED

Name:

Search | Disconnect

Stop Program | Shutdown

Programs | Samples | Console | Settings | Sensors | Motors | Wifi | Bluetooth | Tools | Images

File	Size	Delete
DriveForward.jar	1630	☐
DriveForward.out	47	☐
DriveForwardAndStop.jar	2397	☐
DriveForwardAndStop.out	47	☐
HelloWorld.jar	807	☐
HelloWorld.out	14	☐

Delete Files | Upload file | Download file | Run program | Play Sound File | Format | Set Default

Delete Files

Upload file

Download file

Run program

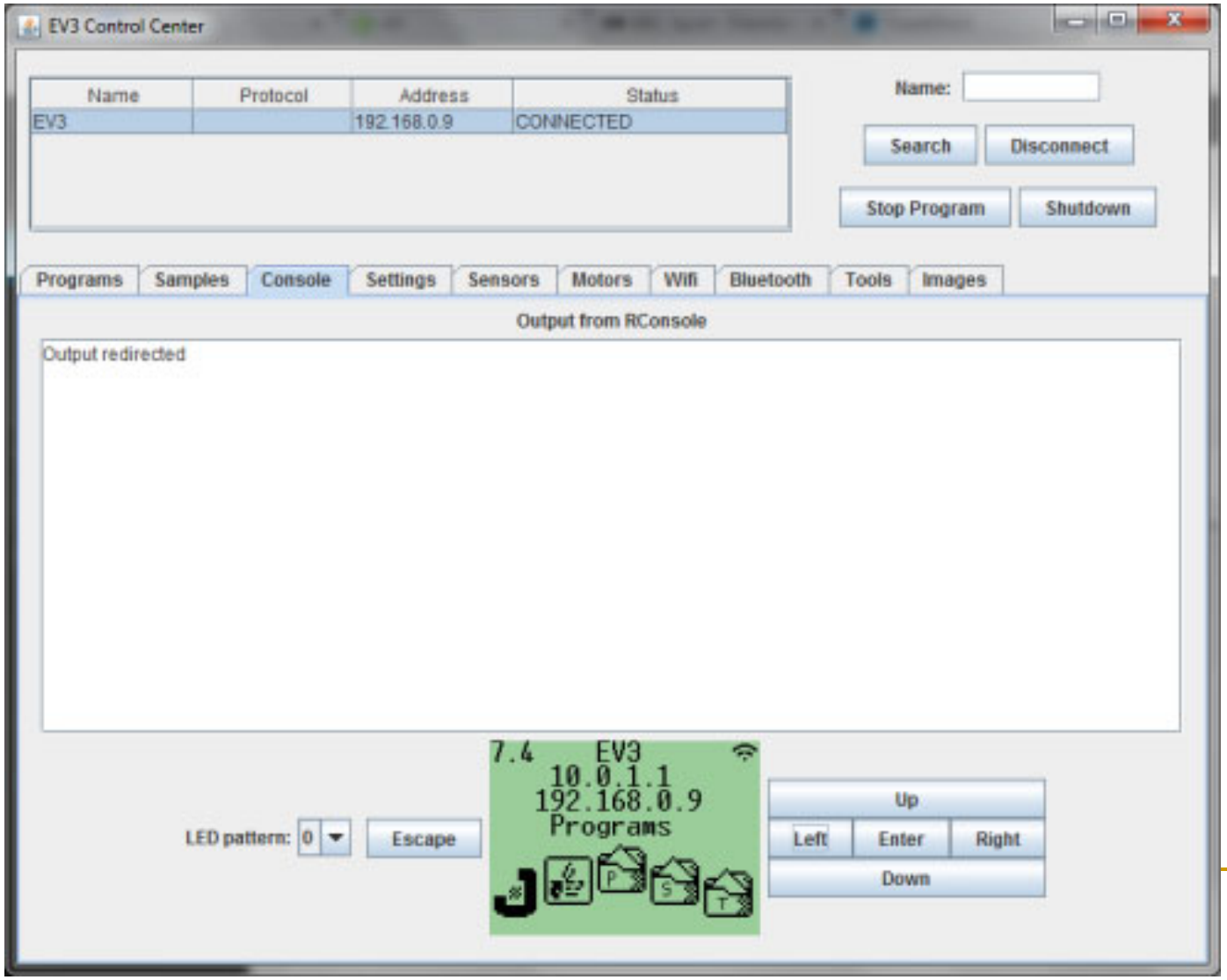
Play Sound File

Format

Set Default

Control Center 4

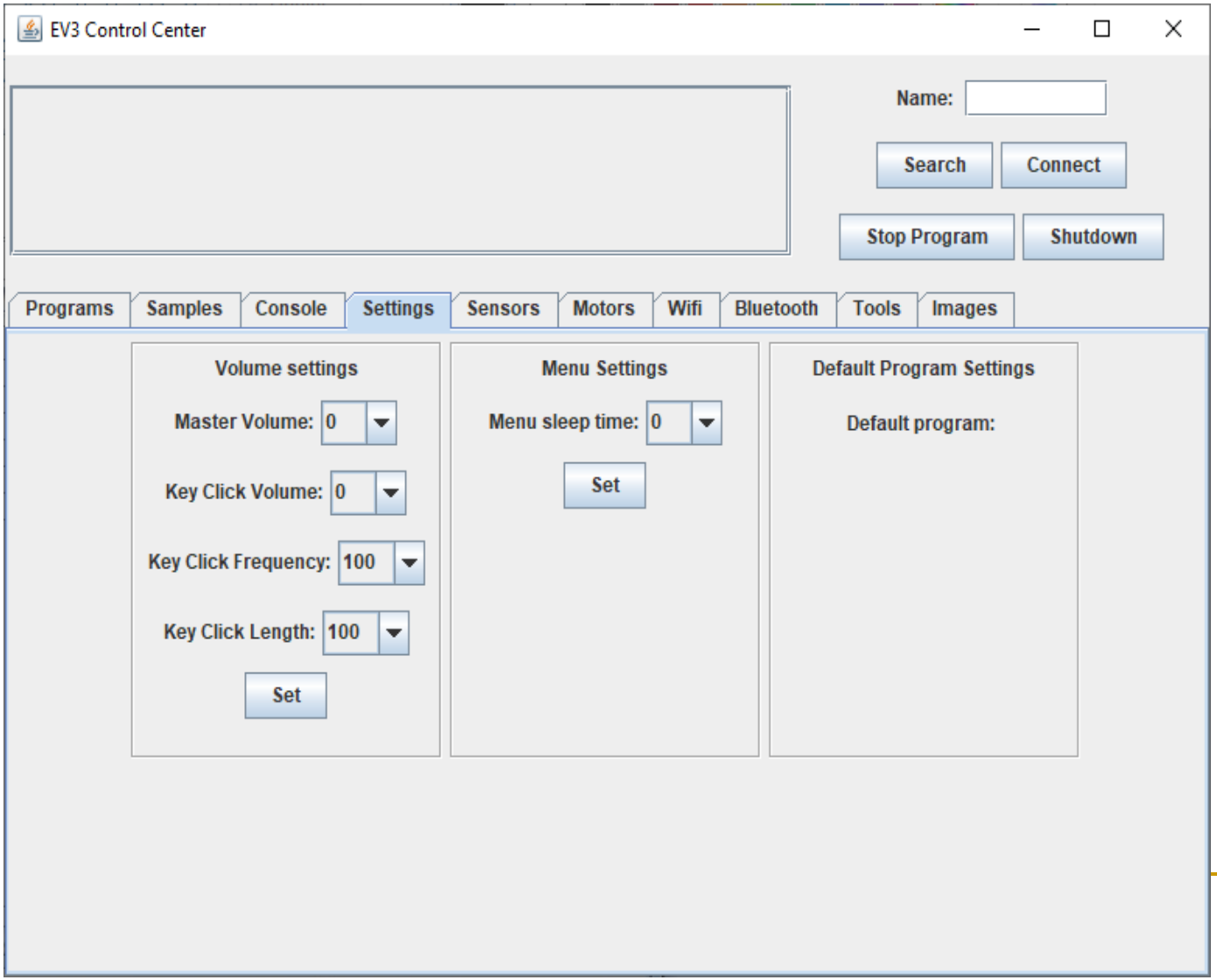
- Console



- View the output on the console
- GUI buttons act like simulated physical buttons
- Live screen of EV3 brick
- Best part of Control Center: **interaction and remote control**

Control Center 5

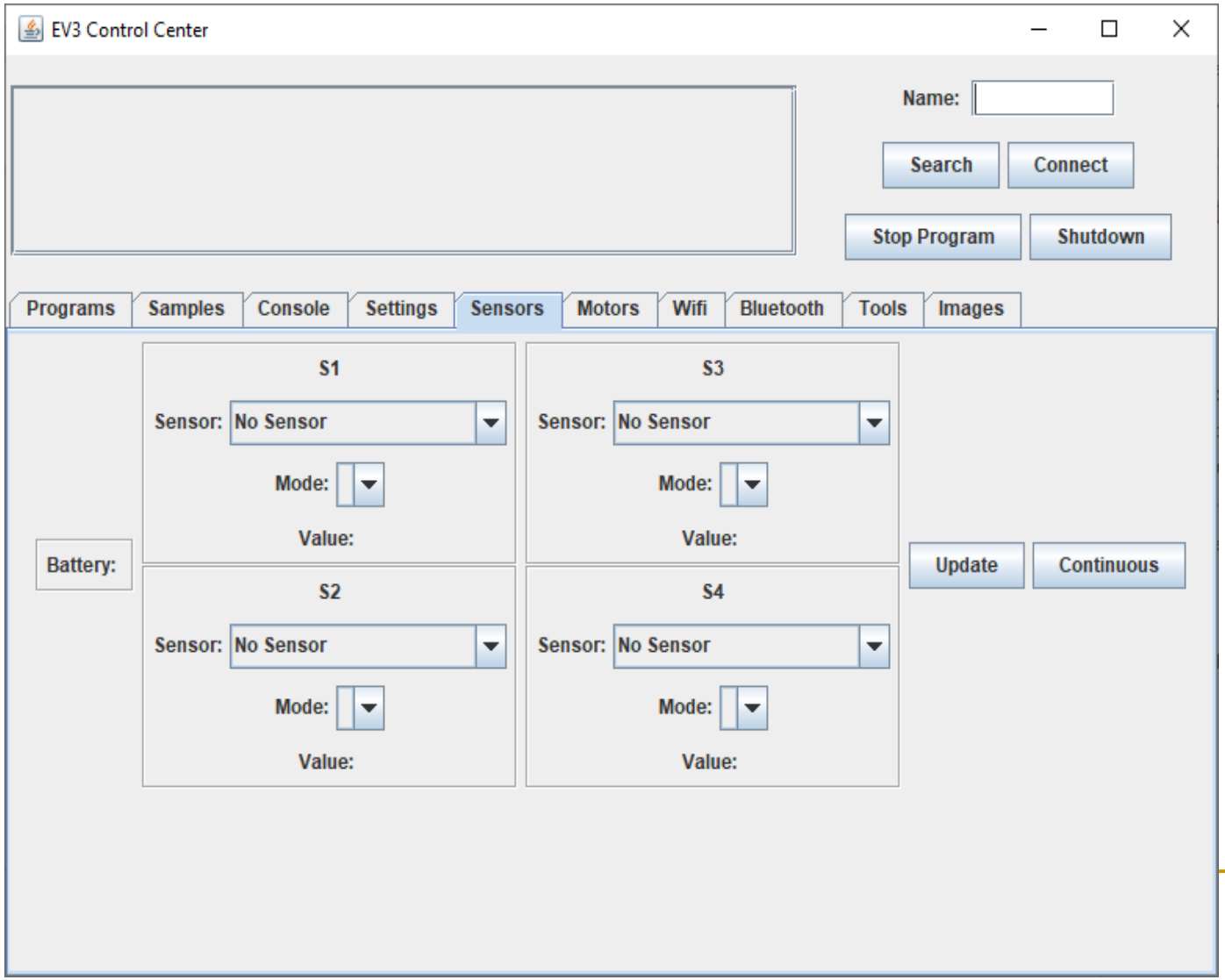
- Settings



- Volume settings
- Automatically power off after the interval of inactivity

Control Center 6

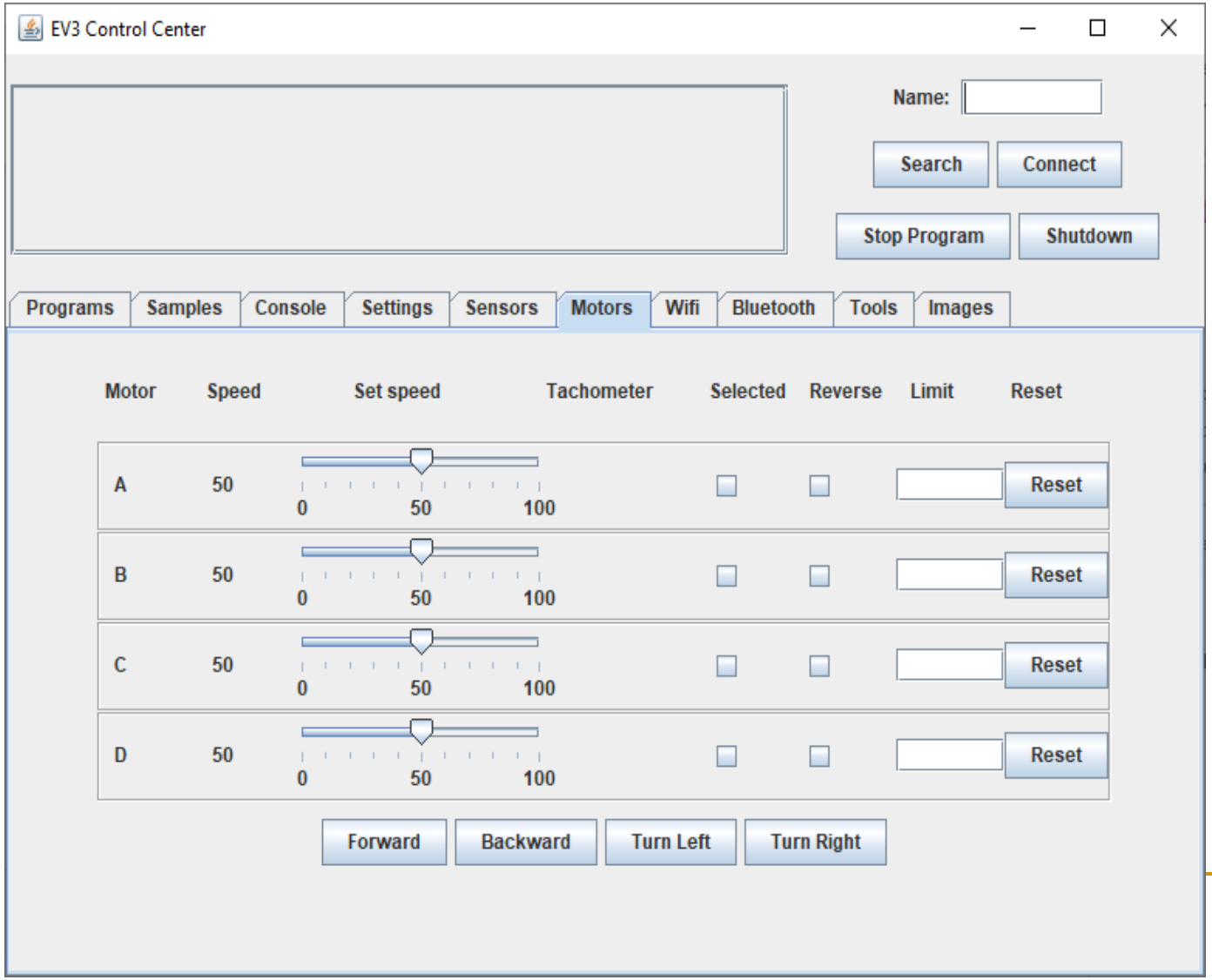
- Sensors



- Battery level
- Real time sensor readings
- Manual update or continuous update
- Sensor + mode → measurements

Control Center 7

- Motors



- Control motors without writing and uploading code to the EV3
- You need to hold down the button, to keep the motor rotating
- You can select two motors
- Limit fields give the max amount of rotation

leJOS APIs 1

- There are basically four main packages in leJOS
  - lejos.hardware.*
    - lejos.hardware.sensor
    - lejos.hardware.motor
    - lejos.hardware.lcd
  - lejos.remote.*
  - lejos.robotics.*
    - lejos.robotics.navigation
    - lejos.robotics.subsumption
  - lejos.utility
- Class files: https://lejos.sourceforge.io/ev3/docs/index.html?help-doc.html
- Source: https://sourceforge.net/p/lejos/ev3/code/ci/master/tree/ev3classes/src/lejos/

leJOS APIs 2

- lejos.hardware.sensor and lejos.hardware.motor are introduced in previous chapter
  - Examples:
    - https://ev3dev-lang-java.github.io/docs/support/fundamentals/sensors/index.html#exteroceptors-sensors
    - https://ev3dev-lang-java.github.io/docs/support/fundamentals/actuators/index.html
- lejos.hardware.ev3
  - This class represents the local instance of an EV3 device. It can be used to obtain access to the various system resources.

leJOS APIs 3

- lejos.hardware.lcd
  - Example:

```
GraphicsLCD lcd = LocalEV3.get().getGraphicsLCD();
lcd.drawLine(5, 5, 60, 60);
lcd.setColor(Color.BLACK);
lcd.drawString("Test", 50,50, 0);

Delay.msDelay(5000);
```

- lejos.hardware.battery
  - Please reference Assignment 00
- lejos.hardware.sound
  - You can upload your own WAV files to play

leJOS Examples

- Examples from installation
  - Samples are optional in the installation: suggested
- Examples from ev3dev
  - https://ev3dev-lang-java.github.io/docs/support/index.html#fundamentals-of-robotics
  - https://github.com/ev3dev-lang-java/examples
    - The imported packages should be changed, but the majority part is the same as leJOS
- Examples in the textbook:
  - http://variantpress.com/books/maximum-lego-mindstorms-ev3/
    - Choose the “Downloads” tab